

Internship Work Update Report

Submitted to

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1. Overview of Internship Tasks and Responsibilities

During my internship, I worked on a wide range of technical tasks spanning e-commerce platform management, front-end and mobile UI development, API integration, cloud technologies, automation scripting, software testing, and documentation. My responsibilities included performing OpenCart migration from version 3 to 4.1, building responsive web and mobile UIs by adapting company dashboard views, modifying service request interfaces, and integrating API-based location data for Walton service centers, plazas, and smart zones. I also developed full-stack applications including URL shorteners and QR code generators using PHP, MySQL, Python, and Node.js, worked with Docker and Kubernetes for containerized deployments, conducted quality assurance testing on the Walton Procurement web portal, managed version control using Git and GitHub, and prepared comprehensive industrial attachment documentation summarizing all technical contributions.

2. Main Objectives

The main objectives of this internship were to gain practical experience in full-stack web development, e-commerce platform management (OpenCart), responsive UI/UX design for both desktop and mobile interfaces, API integration and mapping, cloud infrastructure and containerization (Docker, Kubernetes, AWS), automation scripting, software testing and quality assurance, version control using Git, and professional documentation of technical work performed during the attachment period.

3. Detailed Work Log

DATE	WORK	WORK HOUR
02-05-26	Orientation and initial task assessment.	(9AM-6PM)
03-05-26	I installed OpenCart, explored their OpenCart account website and wantonbd.com , created a custom page in OpenCart, performed full CRUD operations (Create, Read, Update, Delete), and tested various extensions and modules.	(9AM-6PM)
04-05-26	I conducted a technical comparison between OpenCart 4.0 and the Walton-New account modules, evaluating how extensions like Magic Zoom™, SEO modules, and specialized landing pages enhance functionality compared to default setups. I also performed a comprehensive bug audit of the Walton website's Water Heater (Geyser) section, identifying critical issues such as mixed-language content, broken capacity filters, and misaligned technical specification tables.	(9AM-6PM)

	My analysis demonstrated that while specific modules significantly improve administrative efficiency and user engagement, the live site requires targeted layout and data consistency fixes to provide a professional user experience.	
05-05-26	A comprehensive technical comparison was conducted between the standard OpenCart 4.0 system and a module-enhanced version, specifically focusing on how extensions like Magic Zoom™, SEO-friendly URLs, and custom landing pages significantly improve user engagement and site navigation. The evaluation also included a compatibility audit for OpenCart version 4.1.0.3, which identified that many legacy modules are no longer natively supported due to architectural shifts toward event-driven systems and stricter coding standards. This work demonstrated that while advanced modules are essential for a professional e-commerce experience, manual code updates are often required to maintain functionality during platform upgrades.	(9AM-6PM)
06-05-26	Expertise was gained in version control by mastering essential Git and GitHub workflows, including account configuration and repository management. Practical skills were developed in using commands for staging, committing, and pushing code, alongside more advanced operations like stashing changes, managing branches, and performing resets to maintain clean project histories. This training focused on ensuring efficient collaboration and code integrity through the use of pull requests, merge tools, and .gitignore configurations to streamline the development lifecycle.	(9AM-6PM)
07-05-26	Learned how to use Docker by first reviewing our instructor's project files and setting up Docker on our laptops. We moved the project into our local environment, ran the containers, and studied the code to understand how everything works together. For our assignment, we successfully added 100,000 random contact records to the database, created a Redis container for	(9AM-6PM)

	faster data handling, and integrated Redis to display customer information efficiently. This hands-on process helped us see exactly how to manage large amounts of data using modern web tools.	
09-05-26	I completed a technical project focused on containerized application deployment and cloud infrastructure. First, I developed and containerized a multi-service PHP customer management application and a Flask web-scraping tool using Docker and Docker Hub. I then orchestrated the architecture including MySQL, PHP-Apache, Redis, and phpMyAdmin by writing custom configuration files to deploy and manage the services seamlessly on a local Kubernetes cluster. Finally, I documented this practical work alongside a comprehensive guide to core Amazon Web Services (AWS) solutions, demonstrating a solid foundational understanding of scalable cloud infrastructure.	(9AM-6PM)
10-05-26	I completed a technical project focused on API development and URL shortener utilities. First, I developed a secure backend Node.js REST API featuring token-based user authentication (JWT) and a CRUD-based task management system, testing the endpoints extensively with Thunder Client. Alongside this, I built a functional web application for URL shortening and QR code generation, which I uploaded to GitHub. Finally, I documented this practical implementation with an analytical business guide on how enterprises leverage URL shorteners for data tracking and scalability, specifically outlining a modern, highly available serverless architecture using AWS Lambda and DynamoDB.	(9AM-6PM)
11-05-26	Developed an automation script using Python inside a Google Colab notebook. I successfully built a functional workflow that dynamically processes lengthy links into compact web addresses while simultaneously generating clean, high-resolution QR codes mapped to those specific destinations. Working directly	(9AM-6PM)

	within the cloud-based Colab runtime allowed me to efficiently test and validate the script execution, ensuring that all generated QR codes seamlessly route to their targeted links without errors.	
12-05-26	Executed a rigorous software testing and quality assurance review on the Walton Procurement web portal. Focusing heavily on UI and UX design standards, I evaluated the registration modules, vendor dashboards, and company profiling interfaces to pinpoint functional glitches and design ambiguities. I documented these findings in a formal QA report, offering actionable solutions to fix repetitive logout buttons, correct faulty input validations, and improve workspace navigation for a more seamless and professional user experience.	(9AM-6PM)
13-05-26	Continued development and testing of the Python-based URL shortener and QR code generator inside Google Colab. Refined code execution workflows and verified script stability through multiple test runs with real-world URLs.	(9AM-6PM)
14-05-26	Integrated the Bitly API into my Python script within Google Colab to automate link management. I configured secure token authentication using the pyshorteners library to dynamically compress long Google Colab URLs into short Bitly links. Additionally, I embedded the qrcode library to automatically generate and save corresponding QR codes for the shortened links, verifying the complete generation and redirection workflow directly from the Colab interface.	(9AM-6PM)
16-05-26	I spent setting up a full-stack environment in XAMPP to build an independent URL shortener and QR code system. On the backend, I constructed a MySQL database to securely log link mappings. For the presentation layer, I scripted a dynamic frontend using PHP that captures long web addresses, saves them to the database, and processes the shortened targets alongside their corresponding	(9AM-6PM)

	scannable QR codes for immediate display.	
18-05-26	Concluding a two-day full-stack development sprint, I finalized a custom QR code generator web application running on a local XAMPP environment. Building upon the MySQL database structures initialized yesterday, I successfully implemented a complete user management portal today containing login, signup, and password reset functionalities. The finished platform leverages Google's built-in generation link to handle real-time code processing and features a custom history section where users can securely view all their saved QR code logs.	(9AM-6PM)
19-05-26	Spent time studying the architecture and practical applications of URL shorteners. I explored how these services function as a backend system, focusing on their ability to map long, complex URLs to unique, compact aliases using encoding or hashing techniques. Beyond the system design aspects, I also looked into their utility for content creators, specifically how shortening tools help manage links in video descriptions, improve social media shareability, and allow for the tracking of click-through analytics.	(9AM-6PM)
20 -05-26 to 23-05-26	Focused on a store locator project, I attempted an alternative implementation using the Google Maps API ; however, this approach encountered significant performance hurdles, specifically failing to fetch external live details.	(9AM-6PM)
24-05-26	Built a standalone HTML file containing the "Basic Information" section (with fields like name, mobile, address, district, product, service center, brand, barcode, problem description, and optional comments), then inserted it into the existing company file at the appropriate location preserving the original design, Tailwind CSS styling, navigation menu, and status bar, so the form now functions seamlessly as part of the company's webpage.	(9AM-6PM)

02-06-26	Performed an OpenCart migration from version 3 to 4.1, copied the technical specification's HTML code, and submitted a complete report by creating a brand new folder containing all the product HTML files.	(9.00 AM - 6.00 PM)
03-06-26 to 8-06-26	Built a mobile UI by adapting the company's existing desktop dashboard views reorganizing key metrics like target overview, policy overview, current month sales, and financial year performance (asset, profit, receivable, inventory, litigation, etc.) and made the entire interface fully mobile responsive for optimal viewing on smaller screens.	(9.00 AM-6.00 PM)
9-06-26	Made several modifications to the service request UI on the company's main website, improving the layout and user experience by refining form fields, updating styling, and ensuring better responsiveness across devices.	(9.00 AM-6.00 PM)
10-06-26	Studied the provided API containing division, district, and area data for Walton service centers, Walton Plaza, and smart zones, and attempted to load the API onto a map to display the different geographic locations accordingly.	(9.00 AM-6.00 PM)
11-06-26	Attempted to install a new extension in OpenCart named Blueberry . This extension was supposed to add some useful features to the e-commerce platform.	(9.00 AM-6.00 PM)
13-06-26 to 14-06-26	Prepared the complete industrial attachment document that needs to be submitted, including all required sections such as company overview, task summaries (API integration, UI modifications, OpenCart migration, mobile responsive design, and service request updates), technical specifications, and supporting details aligned with the work performed during the attachment period.	(9.00AM-6.00 PM)

4. Conclusion

Overall, this internship helped me understand real workplace responsibilities and strengthened my confidence in handling web development and documentation-related tasks.